

popquiz levers, machines, equilibrium - bonus

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:45pm

Quiz Instructions

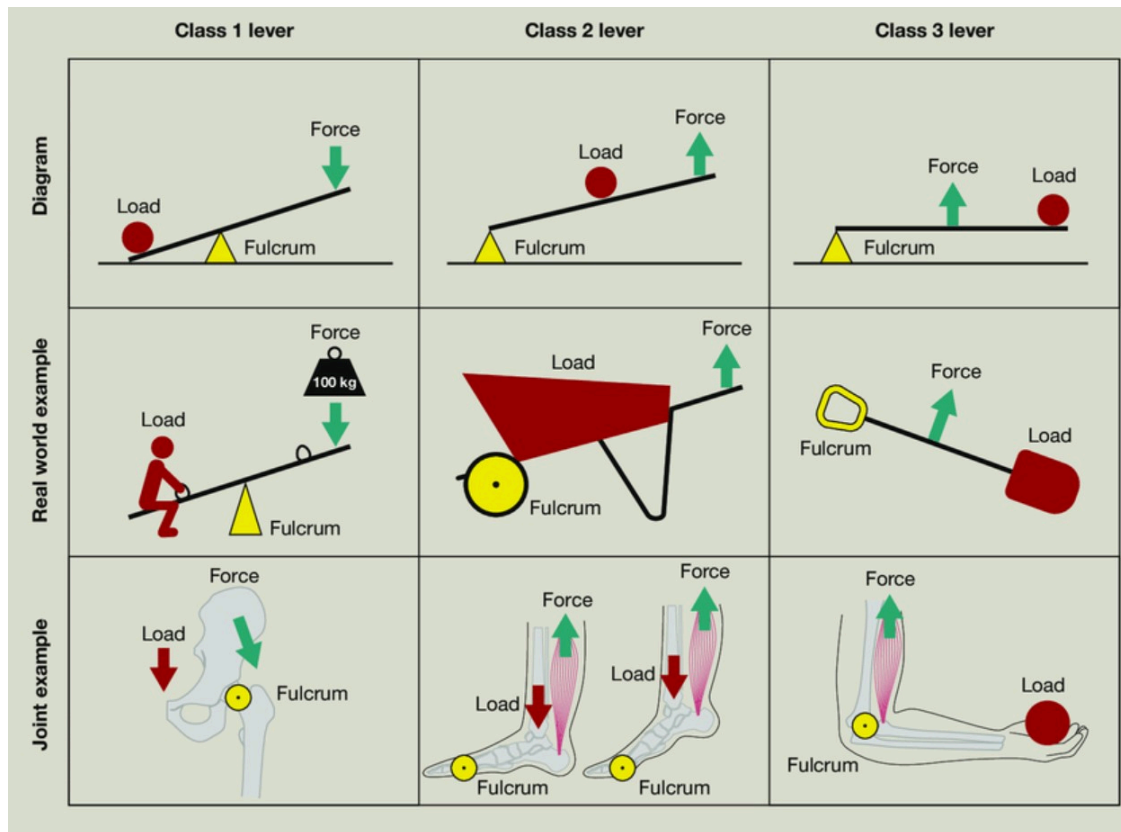
class 1 levers can amplify force or distance

class 2 levers only amplify force

class 3 levers only amplify distance.

to amplify force: distance fulcrum-effort > distance fulcrum-load

to amplify distance: the opposite.

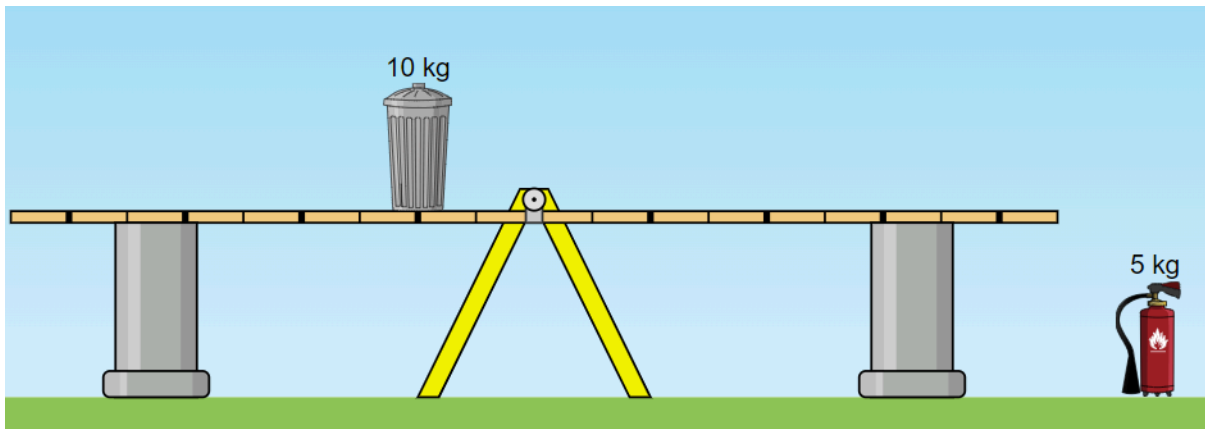


Question 1 10 pts

https://phet.colorado.edu/sims/html/balancing-act/latest/balancing-act_en.html 

(https://phet.colorado.edu/sims/html/balancing-act/latest/balancing-act_en.html)

You use this simulation to help you.

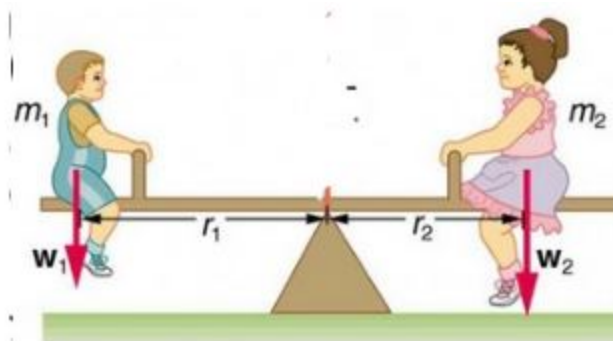


10kg is placed at the position 2 units of distance from the fulcrum. Where should we placed the 5 kg on the otherside? You can verify with the App by removing the poles.

- ☐ impossible to balance out
- ☐ 6 units from the fulcrum
- ☐ 4 units from the fulcrum
- ☐ 2 units from the fulcrum



Question 2 14.5 pts



Little boy has a mass of 100 pounds and sit at one end of the Seesaw. Little girls has a mass of 150pounds. The Seesaw has a total length of 8 meters (!!! the total length is 8m so little boys is m from the fulcrum). How far from the fulcrum is little girl?

hint: use $\text{weight}_1 \times r_1 = \text{weight}_2 \times r_2$ Don't convert. Its an equality. as long as you have pounds on both sides, you don't need to convert to newtons.

- ☐ 2.7m (about 3m)



5,3m (about 5m_



6m



4m



Question 3 14.5 pts

Some levers amplify motion and some amplify force.

Our forearm is a lever. The elbow is the fulcrum. This machine :

See the header of this popquiz.



is not a machine



I missed the class



amplifies force



amplifies motion



Question 4 14.5 pts



What is true about this lever?

hint: see the header of this popquiz



Its a first class lever and it amplifies effort



Its a second class level and amplifies effort



Its a third class class and amplifies motion

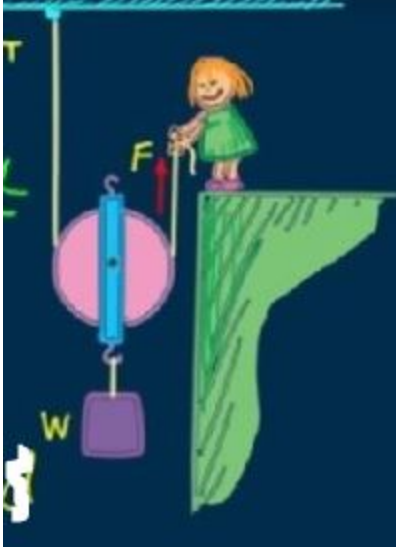


Its a second class level and amplifies motion



Question 5 14.5 pts

Here is little Suzie and her machine. not helping you



She is lifting a 10pounds weight (load). The load (10 pounds) moves by 2 yards. What is true.



her effort is 5 pounds, she pulls 2 yards



her effort is 10 pounds, she pulls 4 yards



her effort is 10 pounds, she pulls 4 yards. dah



her effort is 5 pounds, she pulls 4 yards



I spaced out during class



Question 6 14.5 pts

Lets go back to first question. Consider the board (planck) only. Its in equilibrium and acted by w_1 , w_2 and the normal from the fulcrum. What is the Normal force?

hint: the normal is from the fulcrum (triangle) to the planck. so w_1 and w_2 are down. N is up. Use static equilibrium (sum up = sum down)



100 N

☐

250 N

☐

150 N

☐

0

☐

Question 7 10 pts

Use my slide about levers and machine. A class 3 lever .. What is true?

☐

has the effort between the fulcrum and the load

☐

has the fulcrum between the effort and the load

☐

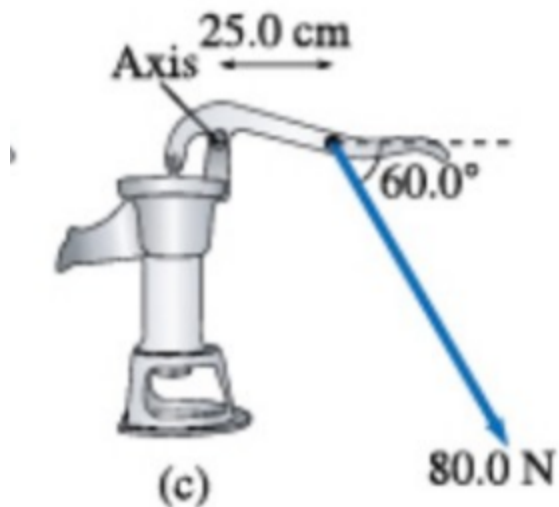
always amplifies force

☐

has the load between the fulcrum and the effort

☐

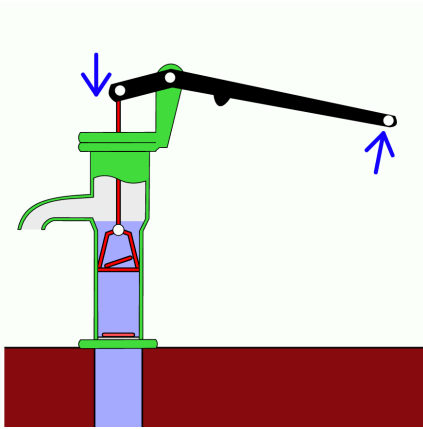
Question 8 14 pts



What is the torque produced by the 80N force (in magnitude). Shown here?

Use the perpendicular component of the force. convert cm to m

Its a water pump)

☐

No answer text provided.

☐

about 20N..

☐

about 10 N.m

☐

about 17N.m

Not saved

Submit Quiz